

JKBOSE 2026

CLASS 11TH

#XUV 

Bhaichava
ON
TOP

ORGANIC CHEMISTRY:

SOME BASIC PRINCIPLES AND TECHNIQUES

LECTURE-02

CHEMISTRY

VISHAL SIR



JK Bole Wallah

Topics To Covered

- 1 Three Dimensional Representation of Organic Molecules**
- 2 Classification of Organic Compounds**
- 3 Functional Group**
- 4 Homologous Series**



OR
COMPLETE
STRUCTURAL
FORMULA.

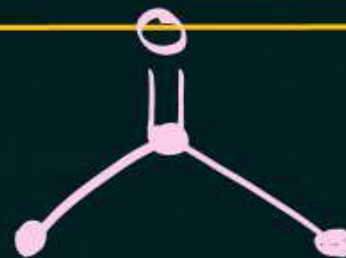
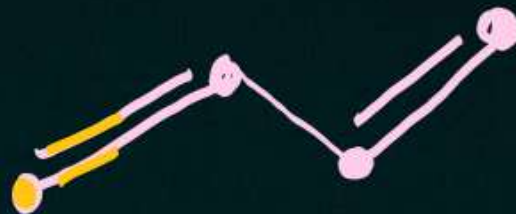
Condensed
formula.

Bond line notation

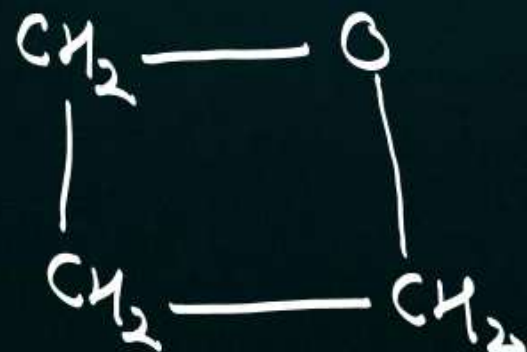
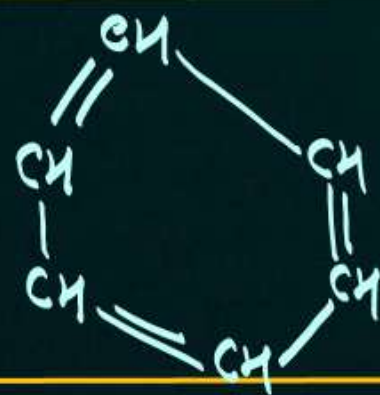
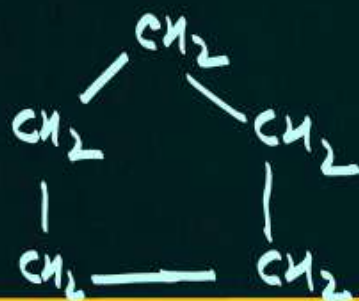
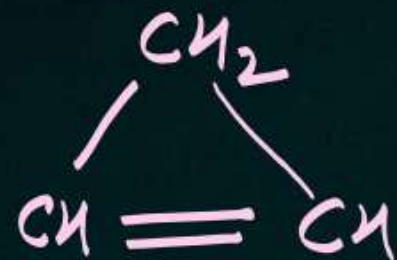
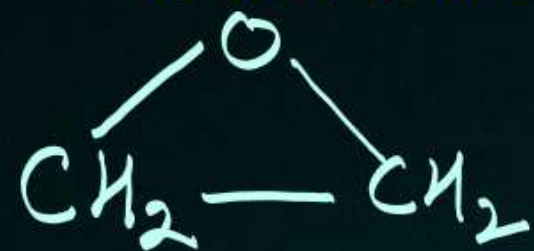
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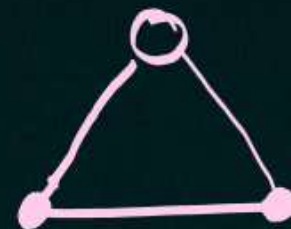
BOND LINE NOTATION.



CONDENSED.



BOND LINE NOTATION.



ORGANIC CHEMISTRY. IS JUST LIKE MATHS.

We have practice it again and again.

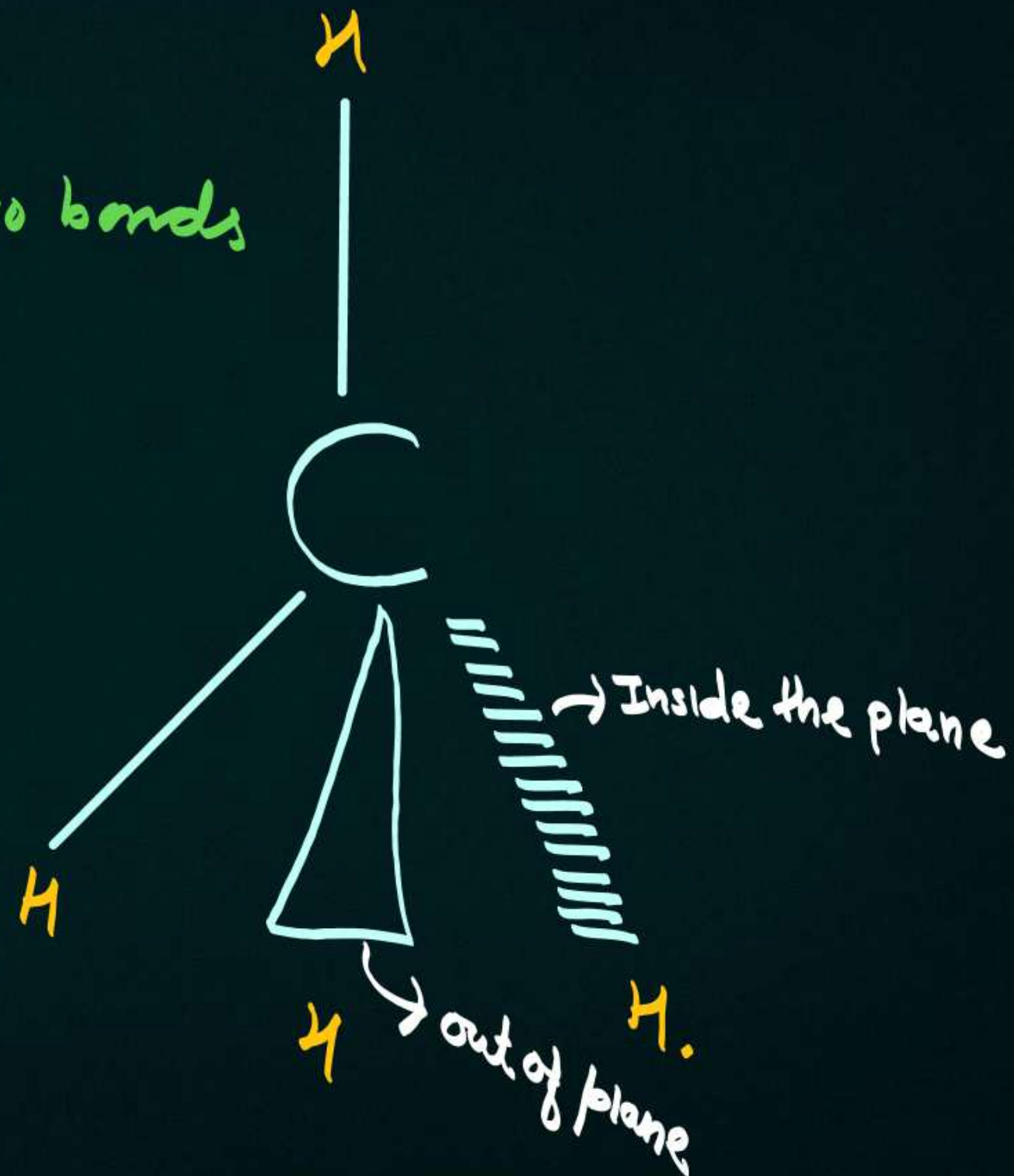
Three Dimensional Representation of Organic Molecules

e.g. CH_4 . As we know CH_4 is sp^3 hybridised and has tetrahedral geometry. i.e. Two bonds are present on the plane and other two bonds are not present on a plane.

- Normal line (—) means bond is present on the plane

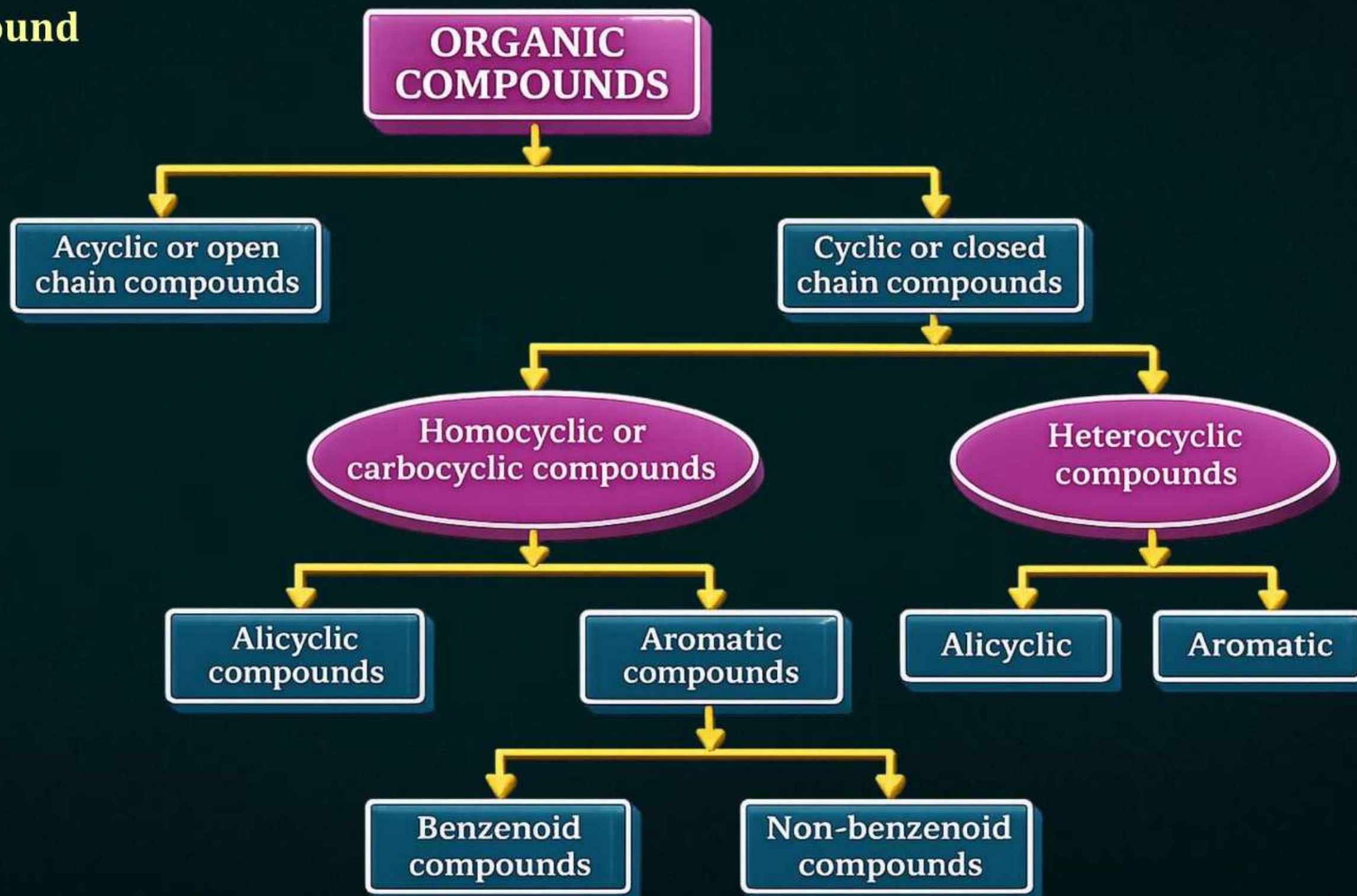
- Wedge line (▴) means bond is towards the observer.

- Dashed line (⋯) means bond is away from the observer



Classification of Organic Compounds

Thus, organic compound
May be classified as:



Classification of Organic Compounds



ORGANIC COMPOUNDS.

OPEN CHAIN
OR
ACYCLIC COMPOUNDS.



CYCLIC OR
CLOSED CHAIN
COMPOUNDS



HOMOCYCLIC

HETEROCYCLIC

ALICYCLIC



AROMATIC.



Benzeneoid

Non-benzeneoid.



ALICYCLIC

AROMATIC

Classification of Organic Compounds

Organic compounds are of two types.

- A) OPEN CHAIN
OR
ACYCLIC compounds.
- B) CLOSED CHAIN
OR
CYCLIC compounds.

A) OPEN CHAIN OR ACYCLIC COMPOUNDS

These organic compounds are also called as aliphatic compounds, in these compounds chain is either straight or branched.

e.g. CH_3-CH_3 , $\text{CH}_2=\text{CH}_2$, $\text{CH}_3-\text{CH}_2-\text{CH}_3$.



B) CLOSED CHAIN OR CYCLIC COMPOUNDS.

In these organic compounds atoms are arranged in such a way that ring formation takes place.

CYCLIC compounds are further of two types.
1) HOMOCYCLIC 2) HETEROCYCLIC.

1) HOMOCYCLIC COMPOUNDS.

Cyclic compounds in which only C & H is present. These are of two types. i) ALICYCLIC ii) AROMATIC.

i) ALICYCLIC $\rightarrow$ Homocyclic compounds that resembles with their parent open chain compounds in properties.



Classification of Organic Compounds

ii) AROMATIC COMPOUNDS $\rightarrow$ cyclic compounds that have pleasant smell or stabilised by aromaticity. These are further of two types. a) Benzenoid b) Non-benzenoid.

a) BENZENOID. Aromatic compounds $\rightarrow$

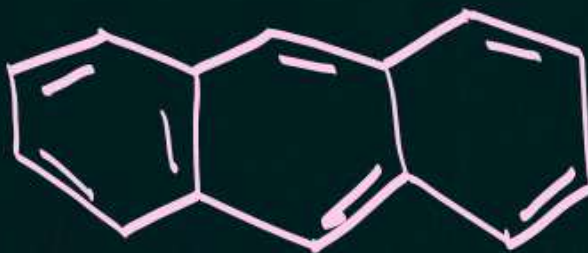
Aromatic compounds having benzene ring are called benzenoid compounds. e.g.



Benzene



Naphthalene



Anthracene

b) NON-BENZENOID $\rightarrow$

Aromatic compounds that do not have benzene ring



Cyclopropenium ion



Cyclopentadienyl anion

Classification of Organic Compounds

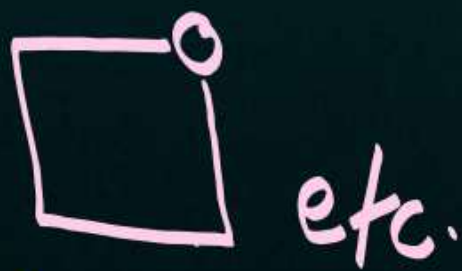


2) HETEROCYCLIC COMPOUNDS.

Cyclic compounds having heteroatom (O & N family) other than C & H. These are of two types a) ALICYCLIC

a) ALICYCLIC ⇒

Heterocyclic compounds that resemble in properties with their parent acyclic compound e.g.



etc.

b) AROMATIC.

b) Aromatic Heterocyclic compounds.

Aromatic compounds containing heteroatom are called aromatic heterocyclic compounds.



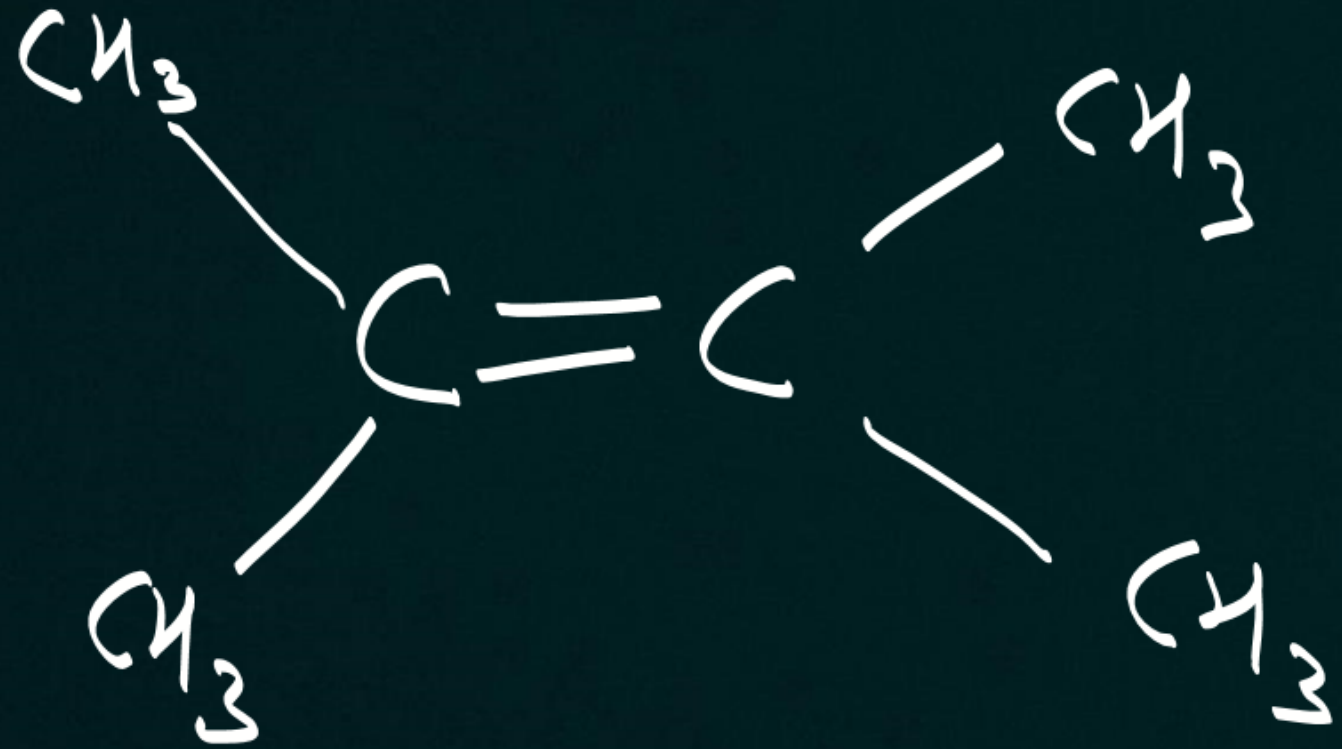
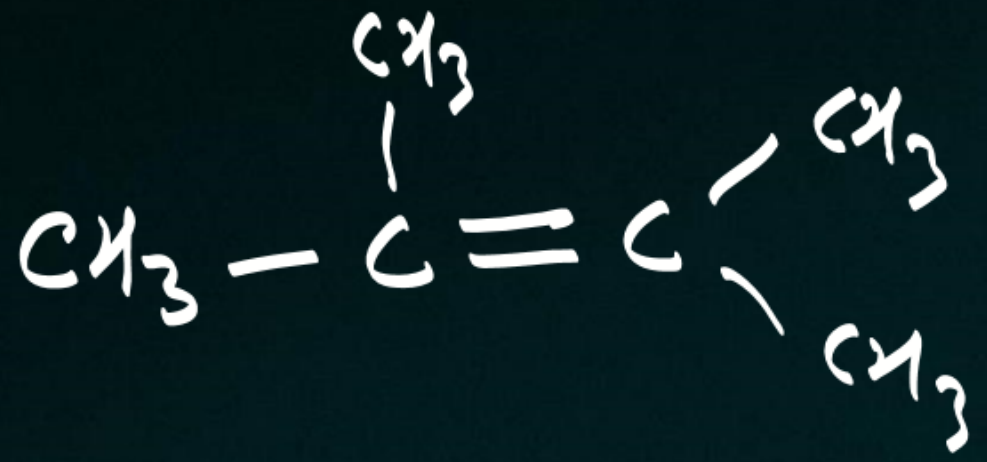
Furan



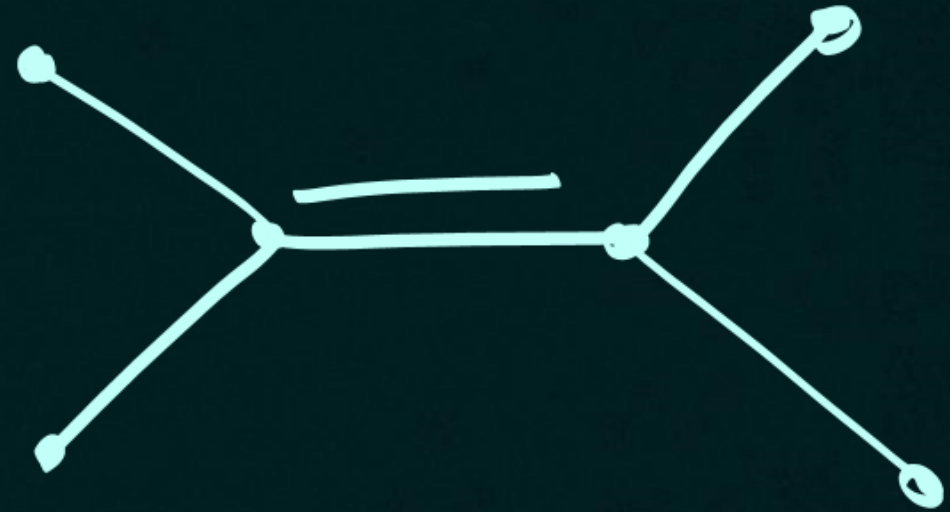
Thiophene



Pyrrole.



=



QUESTION



H.W

Expand each of the following bond-line formulas to show all the atoms including carbon and hydrogen.



QUESTION



The triple bond consists of

- 1 Three sigma bonds 
- 2 Three Pi bonds 
- 3 Two sigma bonds and one Pi bond 
- 4 One sigma and two Pi bonds 



QUESTION

Carbon in methane involves the hybridisation:

1 sp^3

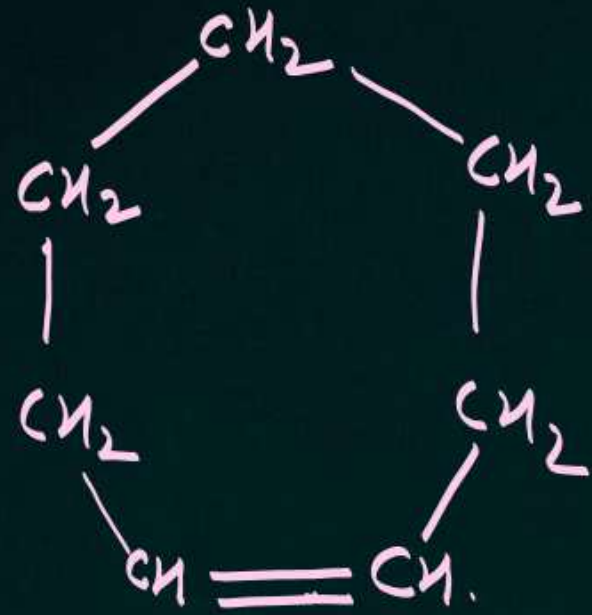
2 sp^2

3 sp

4 None of these



Home Work.



MEERA JIGAR PADA KARO.

Thank
You